

Cultural Aspects of Global Requirements Engineering:

An Empirical Chinese-German Case Study

Patricia Shiroma Brockmann
Computer Science Department
Ohm University of Applied Sciences
Nürnberg, Germany
Patricia.Brockmann@ohm-hochschule.de

Thomas Thaumüller
Computer Science Department
Ohm University of Applied Sciences
Nürnberg, Germany
ThaumullerTh41145@ohm-hochschule.de

Abstract— Global software development places extremely high demands on requirements engineering. Learning to combine requirements engineering techniques across national, cultural and language borders presents unique challenges for developers. In this paper, we explore the significant cultural challenges to requirements engineering in global software development. As an empirical case study, we look at the difficulties involved in applying requirements engineering methods to a joint Chinese-German software project.

Keywords- *global, software, requirements, engineering, agile, China, Germany, cultural aspects*

I. INTRODUCTION

The increased prevalence of global software development increases both technical as well as cultural complexity. Requirements engineering, in particular, requires higher levels of communication, self-organization and frequent adaptation to a dynamic environment. In order to successfully requirements engineering methods to global software development, practitioners need to learn to communicate and collaborate effectively with team members from other countries and from other cultures.

In this paper, we will investigate the use of requirements engineering in global software development. As a case study, we investigate the cultural challenges involved in applying agile requirement engineering methods in China. We present empirical experiences encountered during an ongoing joint Chinese-German software development project for a Chinese regional bank.

In Section 2 we discuss the cultural challenges involved in global software development. Section 3 explores the application of agile methods to global software development. Section 4 presents a detailed analysis of the application of agile requirement engineering methods in China and Germany. Section 5 presents our empirical results from an ongoing joint Chinese-German software development project. Section 6 discusses future work.

II. CULTURAL CHALLENGES

As an aide to handling the complexity of cultural challenges, a number of authors have classified cultural differences according to different categories. Hofstede [1] used

sophisticated multivariate statistical methods to analyze data collected from thousands of IBM employees around the world. Hofstede classifies differences of cultural perspectives in terms of five dimensions:

- Power distance
- Collectivism vs. individualism
- Masculine vs. feminine
- Uncertainty avoidance, ambivalence
- Long-term vs. short-term orientation

Trompenaars and Hampden-Turner [2] developed an expanded list based on knowledge gained from their consulting work for large, multinational companies. They distinguish the following categories of cultural differences:

- Universalism vs. particularism
- Individualism vs. communitarianism
- Neutral vs. emotional
- Specific vs. diffuse
- Achievement vs. ascription
- Attitude to time
- Attitude to the environment
- Gender, race, class, religion

For example, Laroche[3] observed that M-time (monochromatic) individuals tend to start and end a review at scheduled times. P-time (polychromatic) people may feel that the review has been cut off abruptly, before they have a chance to express their views. Learning to analyze cultural differences in terms of alternate perspectives of the same situation can help students move beyond cultural stereotypes.

A number of authors have investigated intercultural aspects with a special focus on software engineering. Requirements Engineering can be adversely affected by different views of power distance. Kruchten [4] investigated the impact of intercultural factors on global software development. Kruchten suggested the development of a matrix [factors x practices] to show which intercultural factors can have an effect on certain software development practices.

One concrete example of how cultural factors can impact software development practices was identified by Thanassanki and Corbitt[5]. They investigated the role that power

distance and uncertainty in Thai culture play in Requirements Engineering. They observed that the deep, hierarchical organization levels and long chains of command in Thai organization require additional time for decision making. They found that often the job title of the person making a decision was more important than technical considerations.

III. GLOBAL SOFTWARE DEVELOPMENT WITH AGILE METHODS

Agile methods in software development, such as Extreme Programming [6] and Scrum [7], have proven quite successful in addressing the uncertainty inherent to the dynamic environment of software development. Test-driven development, pair programming, continuous integration and frequent releases were originally designed for use at one single site. Continuous, face-to-face interaction between developers and also with the customer is a core principle of agile methods. The geographical and temporal separation of developer teams and customers makes this higher level of communication extremely difficult.

The suitability of agile methods in globally distributed software development has been investigated by a number of authors. Sureshchandra und Shrinivasavadhani[8] discuss the successful introduction of a distributed agile model in a large software development company in India. Frequent rotation between onsite and offshore teams, as well as video conferencing equipment between the agile sites, aided in distributing business knowledge among teams.

Paasivaara and Lessenius[9] identify communication as one of the biggest challenge in distributed software development. Misunderstandings, especially in requirements engineering, are a common source of problems. Especially the informal communication necessary to build trust and respect among team members can be difficult to establish. Fowler [10] suggests sending “ambassadors” from one site to another. Because these ambassadors are responsible for informal communication, they should be rotated every few months so that they don’t lose contact with home.

What role do intercultural aspects play in acceptance levels of agile methods? Agile methods emphasize constant change and adaptation. Cultures with high uncertainty avoidance paired with long-term orientation may view constantly changing requirements inherent to agile methods as frustrating. Furthermore, the introduction of agile methods can upset hierarchical organizational structures within a company. In a country with high power distance, skipping over established hierarchical levels may result in resistance. Quick, non- bureaucratic decisions made at the operative level may prove completely unrealistic.

IV. AGILE REQUIREMENTS ENGINEERING METHODS IN CHINA AND GERMANY

As part of an empirical case study, we investigate the suitability of agile requirement engineering methods for use in two very different countries, China and Germany. The

reason we suggest using China and Germany is because of the large cultural distance between east Asian and western European cultures.

First, cultural aspects of Germany were analyzed according to Hofstede’s five dimensions of culture[1] and in relation to German cultural heritage as rooted in western European social market economy. For each cultural dimension, a maximum of 100 points is possible.

- Power distance: The relatively low power distance (30 points) can be seen as consistent with a relatively equal distribution of power in social market economy.
- Collectivism vs. individualism: An above average value (62 points) for individualism can also be seen as a preference for individual fulfillment over the good of the whole.
- Masculinity vs. femininity: An relatively high value (61 points) for masculinity reflects high degree of gender differentiation of roles. This is also consistent with a positive appreciation of competitive and assertive behavior
- Uncertainty avoidance, ambivalence: A relatively high value (60 points) for uncertainty avoidance reflects an appreciation for rules, clarity and structured situations, coupled with a low tolerance for ambiguity.
- Long-term orientation: Hofstede found a surprisingly low value (25 points) for long-term orientation in Germany. Respect for tradition and fulfillment of social obligations are characteristic for countries with short-term orientation. Thrift and perseverance are not valued as highly in countries with short-term orientation.

What effect do these cultural dimensions have on the acceptance of agile requirement engineering methods in Germany?

- Power distance: Self-organization and a lack of fixed hierarchical structures, as advocated by methods, are well-suited to countries like Germany with a low power distance.
- Collectivism vs. individualism: Agile methods emphasize the group over the individual, e.g. by advocating pair-programming and joint ownership of code. This can constrain the individualist tendencies of many German programmers.
- Masculine vs. feminism: Cooperation, communication and pair-programming are typical “feminine” characteristics, which may conflict with “masculine” competitive and assertive characteristics valued in Germany.
- Uncertainty avoidance: Unstructured situations encountered in agile methods may lead to a lack of acceptance in societies which value rules, safety and predictability. A low tolerance for ambiguity be advantageous in formulating clear, concise project requirements and in setting priorities of which features are most important.

- Long-term orientation: Frequent iterations of requirements elicitation and short-term sprints in agile methods mesh well with Germany's relative low value for long-term orientation.

Next, cultural aspects of China were analyzed according to Hofstede's five dimensions of culture[1] and in relation to Chinese cultural heritage as rooted in Confucianism.

- Power distance: The very high power distance (80 points) can be seen as having its roots in Confucian philosophy. Respect for elders and a fixed, hierarchical societal structure are traditional values.
- Collectivism vs. individualism: A very low value (20 points) for individualism can also be seen as closely connected to the Confucian precept that the good of the whole society plays a higher role than the good of the individual. This may also be a reflection of decades living in a collectivist society under communist rule.
- Masculinity vs. femininity: An average value (50 points) for masculinity is consistent with a Confucian disapproval of competition and aggressive behavior.
- Uncertainty avoidance, ambivalence: Hofstede found a surprisingly low value (35 points) for uncertainty avoidance. This result seems surprising at first glance, because this does not seem to be consistent with hierarchical Confucianism structures which have a more negative view of change. If we look at Chinese views toward ambivalence, however, this result makes more sense. A low value for uncertainty avoidance is found in cultures with relativistic philosophies and an inhibition towards expressing emotions in public.
- Long-term orientation: China has one of the highest values (98) for long-term orientation. Traditional Confucian values which place the family unit at the center advocate thrift, perseverance and sacrifice for coming generations as opposed to present consumption. This dimension may also have changed since the recent market reforms.

What effect do these cultural dimensions have on the acceptance of agile requirements engineering methods in China?

- Power distance: Self-organization and a lack of fixed hierarchical structures as advocated by agile methods seem in direct conflict to Confucian thought. Verbal communication across all team members, regardless of their roles, does not fit well with a high power distance. In cultures with a high power distance, the role of the person who says something can be more important than the substance of what is being said.
- Collectivism vs. individualism: Agile methods advocate joint ownership of code and team responsibility. This is in keeping with a high value of collectivism.

- Masculine vs. feminism: Cooperation, communication and pair programming fit well with Confucian ideals of group harmony.
- Uncertainty avoidance, ambivalence: Acceptance of ambivalence can hinder agile requirements engineering methods. Clear, concise story cards or verifiable, verbal elicitation from on-site stakeholders are vital to the success of agile methods. Priorities of which features need to be implemented first have to be defined exactly.
- Long-term orientation: Frequent iterations and short-term sprints do not conform to Confucian ideals of long-term orientation. Constant change may be viewed as frustrating to Chinese team members.

In summary, east Asian and western European cultures have quite different views on Hofstede's five dimensions of culture. These cultural differences present significant challenges to successful requirements engineering. Whether agile methods can help or hinder this process will be investigated in the next section.

V. REQUIREMENTS ENGINEERING IN A JOINT CHINESE-GERMAN SOFTWARE PROJECT

The hypothetical analysis presented above is currently being tested as a case study on a practical software development project. The case study involves the requirements engineering, design and development of personnel software for a Chinese regional bank. Team members of the project are from the two countries analyzed above, Germany and China.

Overcoming the language barrier between team members from the two countries is one problem which could be anticipated beforehand. Thus, additional time support for language translation could be accounted for in the planning. Although all of the team members had to overcome some language barriers, they were usually able to communicate adequately in English.

Time and geographical distances were also aspects which could be anticipated beforehand and taken into account in advance planning. Since the constant on-site presence of all stakeholders was not feasible, regular telephone calls and video conferences could be scheduled for frequent iterations of requirements elicitation. E-mail communication often led to more misunderstandings, due to additional ambiguities introduced in using a written language (English) which was foreign to both groups. As a result, e-mail was mainly used for asynchronous communication, to bridge temporal differences.

Overcoming cultural barriers, however, proved to be more formidable. Team members with similar cultural expectations, such as those from different western European countries, had less difficulty working together. The larger the cultural distance between team members, such as those between China and Germany, the more difficulty they had in coordinating their different styles of communication, conflict management and especially in how individual members dealt with criticism and ambiguity. For example, German team

members were often very open and direct in their criticism, which often lead to a “loss of face” among Chinese team members. Chinese team members expressed their requirements in a form considered ambiguous to German team members. Feedback from the unit tests from Chinese team members was often so politely worded in order to avoid direct criticism of an individual, that German team members misinterpreted the message as a positive validation that the results of unit tests were correct.

How can such cultural misunderstandings be minimized? First, we return to Hofstede’s cultural dimensions and offer suggestions to bridging gaps for each aspect.

- **Power distance:** Self-organization and a lack of fixed hierarchical structures prescribed by agile methods may be disturbing in cultures like China with high power distance. One solution is to adapt some of the agile roles, such as “Scrum Master” to better fit hierarchical organizational structures. Instead of a democratically elected scrum master, the person with the highest status in the group would receive this title.
- **Collectivism vs. individualism:** Agile methods advocate joint ownership of code and team responsibility. This can be a problem for some German team members, who have a high level of individualism. Here, team training workshops and group social events may be of help in increasing group cohesiveness.
- **Masculine vs. feminism:** Cooperation, communication and pair-programming advocated by agile methods can be hindered by a high value for masculinity, as in Germany. Masculine values such as competition and assertive behavior can be constructively redirected through conflict management techniques.
- **Uncertainty avoidance, ambivalence:** A high level of ambivalence, as in China, can prove to be a major obstacle to agile requirements engineering methods. German developers, who expect clearly defined project requirements, express puzzlement at not understanding what their Chinese customers want. Frequent communication and cultural awareness training can help increase understanding on both sides.
- **Long-term orientation:** Frequent iterations and short-term sprints may be viewed as frustrating to Chinese team members, who highly value long-term orientation. Here, schooling and communication about the goals and effectiveness of agile methods may be of help in defining the numerous iterations as steps toward an ultimate goal: software that works.

Especially the last two dimensions discussed, uncertainty avoidance and long-term orientation, prove the most challenging in integrating groups with large cultural distance. Fowler’s [10] suggestion to send “ambassadors” from one site to another, or frequent rotation of team members between sites, as advocated by Sureshchandra [8] are two methods which could help in team building and to increase cultural understanding. As both methods require significant

financial resources, they were unfortunately not possible for this project.

Kruchten discusses the use of proxies as one possible solution to bridge gaps between cultures. He defines proxies as “bicoded” individuals, able to operate equally at ease in two different cultures[4]. A typical proxy is someone born and raised in one culture, who has lived and studied for a number of years in a different culture. A proxy who has internalized the values and associated behaviors of a second culture can serve as an important bridge between two cultures.

Foreign students can make an extremely valuable contribution as proxies. Proxies can be exchange students (from China) who spend a number of years studying at a foreign university (Germany). German students, who have spent a year studying or completing an internship abroad (in China), have had the opportunity to learn firsthand about another culture. Such proxies have a deepened understanding of two different languages and cultures and can thus aid in the translation of both language as well as cultural mentality.

We paired groups of “native” with “proxy” team members to form working teams. Proxy team members were vital in translating the differences between “what was said” and “what was meant” between the Chinese and German team members. Proxies were able to translate Chinese cultural concepts to German team members, such as how to avoid “losing face”, or how to “give face” by complementing someone’s work, and thus increase the prestige of all parties involved. Proxies translating communication from the German team members could reword direct criticism, in order to avoid offending Chinese team members. Proxies were also better at interpreting the subtle nuances of requirements elicitation from Chinese stakeholders, which were often overlooked by German developers. These proxies proved to be the most valuable practical aid in learning to bridge cultural differences.

VI. CONCLUSIONS AND FUTURE WORK

Effective requirements engineering in a globally distributed environment requires not only new technical knowledge and tools, but also the ability to understand unfamiliar situations from different cultural perspectives. Analysis and communication of different values for cultural dimensions can help increase awareness and improve the effectiveness of communication between team members. The use of proxies can be an effective method to bridge cultural gaps between team members from different countries. The experiences gained from this project will be applied to a future project

REFERENCES

- [1] Hofstede, G. and Hofstede, G.J., *Cultures and Organizations: Software of the Mind*. McGraw-Hill, 2nd Edition, 2005.
- [2] Trompenaars, F. and Hampden-Turner, C., *Riding the Waves of Culture*, 2nd Edition, McGraw-Hill, 1997.
- [3] Laroche, L. *Managing Cultural Diversity in Technical Profession.*, Butterworth-Heinemann, 2002.
- [4] Kruchten, P., "Analyzing Intercultural Factors Affecting Global Software Development – A Position Paper", *Proceedings of GSD2004 3rd International Workshop on Global Software Development*, IEEE 2006.
- [5] Thanasankit, T. and Corbitt, B. J., "Thai Culture and Communication of Decision Making Processes in Requirements Engineering". *Proceedings of CATAC 2000, Cultural Attitudes Towards Technology and Communication*, Perth, 2000.
- [6] Beck, K., *Extreme Programming Explained: Embrace Change*. Addison-Wesley, 2000.
- [7] Schwaber, K. and Beedle, M., *Agile Software Development with Scrum*. Prentice-Hall, 2002.
- [8] Sureshchandra, K. and Shrinivasavadhani, J., "Adopting Agile in Distributed Development". *IEEE International Conference on Global Software Engineering*, IEEE, 2008.
- [9] Paasivaara, M. and Lassenius, C., "Could Global Software Development Benefit from Agile Methods?" *IEEE International Conference on Global Software Engineering*, IEEE, 2006
- [10] Fowler, M., *Using Agile Software Process with Offshore Development*. (<http://martinfowler.com/articles/agileOffshore.html>), July 18, 2006
- [11] Verkamo, A.,I. et al., "Distributed cross-cultural student software project: A case study". *CSEE&T*, 2005.
- [12] Lago, P., Muccini, H., AliBabar, M., "Developing a Course on Designing Software in Globally Distributed Teams". *IEEE International Conference on Global Software Engineering*, IEEE, 2008.